

SUPPLEMENTAL MATERIAL

Table S1. Average PM_{2.5} at the percentiles categories of gross β -activity: the joint distribution of PM_{2.5} and gross β -activity.

gross β-activity	0 - 10th percentile	10th percentile- median	Median - 90th percentile	90th percentile - max
PM _{2.5} (ug/m ³)	6.09	7.67	9.50	8.20

Table S2. Distribution of annual cause-specific death counts per ZIP code tabulation area in different age groups: 18-65, 65-85, 85+, across years and ZIP code tabulation areas.

Death Count	Mean	Standard deviation	10 th percent tile	25 th percent tile	Median	75 th percent tile	90 th percent tile
18-65 age group (n)							
Cardiovascular disease	3.9	4.7	0	1	2	5	10
Myocardial infarction	0.8	1.3	0	0	0	1	3
Stroke	0.5	0.9	0	0	0	1	2
Non-accidental causes	18.3	19.6	1	4	12	25	46
65-85 age group (n)							
Cardiovascular disease	10.7	11.7	1	2	7	15	26
Myocardial infarction	2.4	3.2	0	0	1	3	6
Stroke	2.3	2.9	0	0	1	3	6
Non-accidental causes	45.0	45.2	3	11	31	65	106
85+ age group (n)							
Cardiovascular disease	11.9	12.9	0	2	8	17	29
Myocardial infarction	1.9	2.6	0	0	1	3	5
Stroke	2.7	3.4	0	0	2	4	7
Non-accidental causes	38.3	40.3	2	8	25	56	93

Figure S1. Spearman correlation between variables included in statistical models. A lighter color represents a weaker correlation between two variables. Blue referred to positive and red means negative correlation. The correlation with a cross can be interpreted as the *p*-value of the Spearman correlation test between these two variables being greater than 0.05.

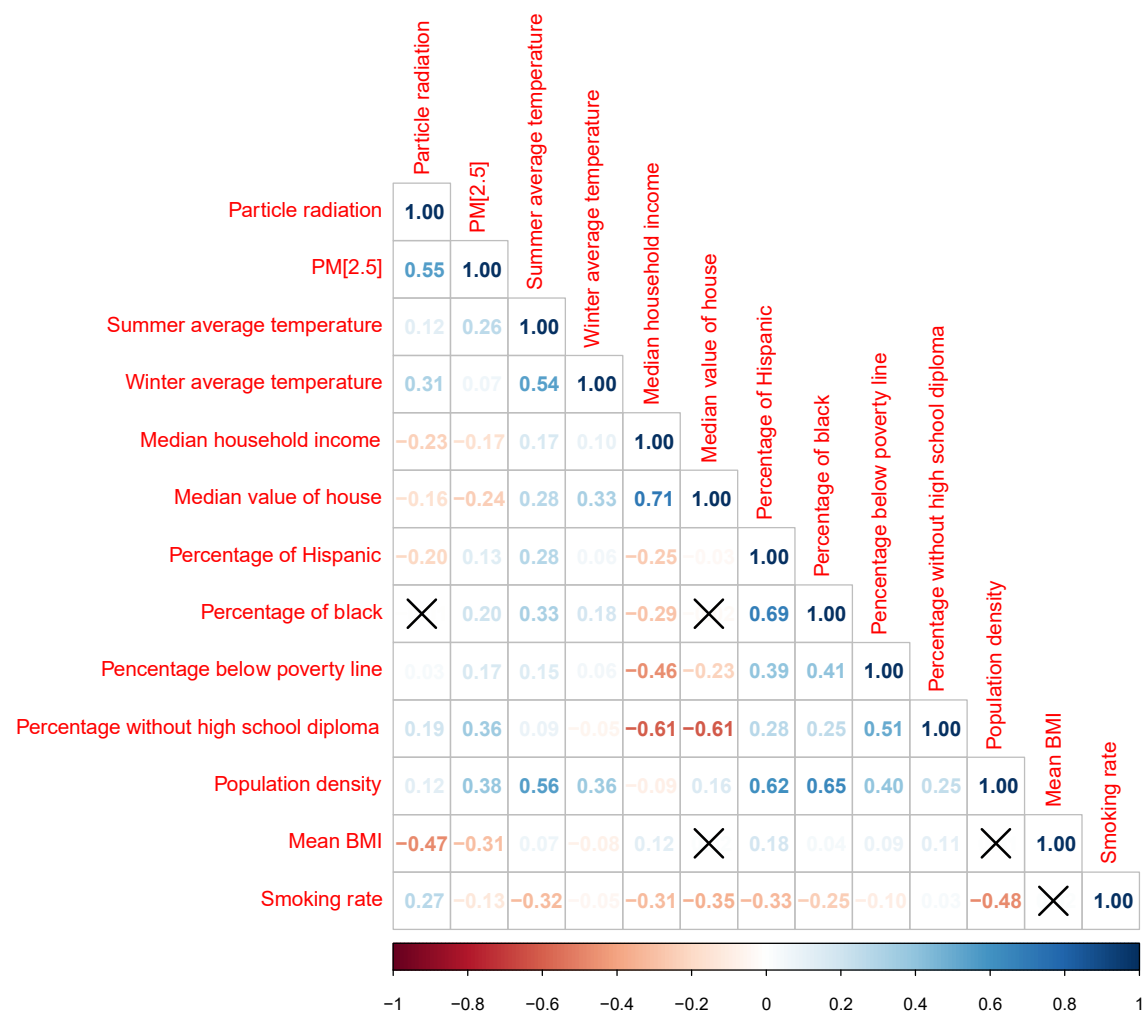


Figure S2. Cause-specific rate ratios and 95% confidence intervals for an interquartile-range (0.055 mBq/m³) increase in gross β -activity controlling PM_{2.5} using both generalized linear mixed model and difference-in-differences approaches in different age groups (18+; 18-65; 65-85; 85+).

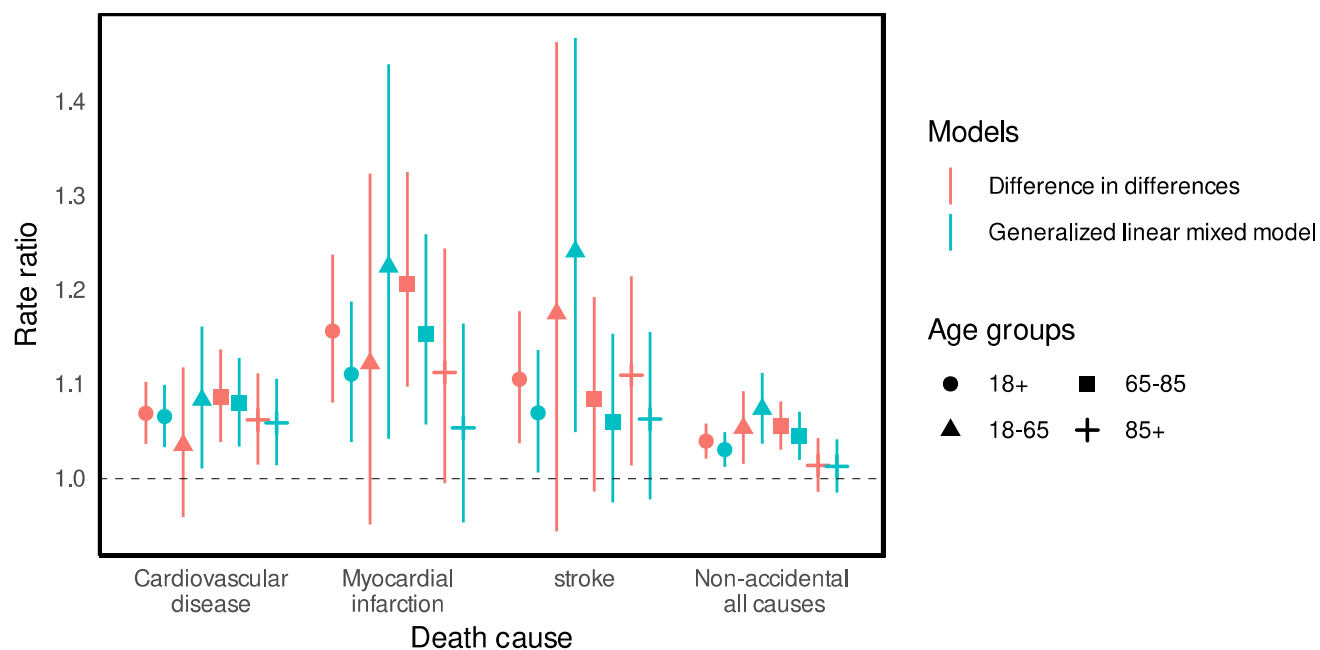


Figure S3. Cause-specific rate ratios and 95% confidence intervals for an interquartile-range ($2.82 \mu\text{g}/\text{m}^3$) increase in $\text{PM}_{2.5}$ in different age groups (18+; 18-65; 65-85; 85+) fitted with both generalized linear mixed model and difference-in-differences approaches.

